

FLOOD-RESILIENT CONNECTIVITY PLANNING

ResiliNet 3D

HACK_MY_060| AI SLOP FINAL BOSS

Malaysia • ASEAN GeoAI Fusion 2026

Introduction

Why connectivity fails exactly when people need it most.



Background: In a flood, mobile sites go down because of power cut-off and backup batteries run out



Theme alignment: Connectivity and network planning. An AI rain forecast drives geospatial models of flooding, road access and radio line of sight.

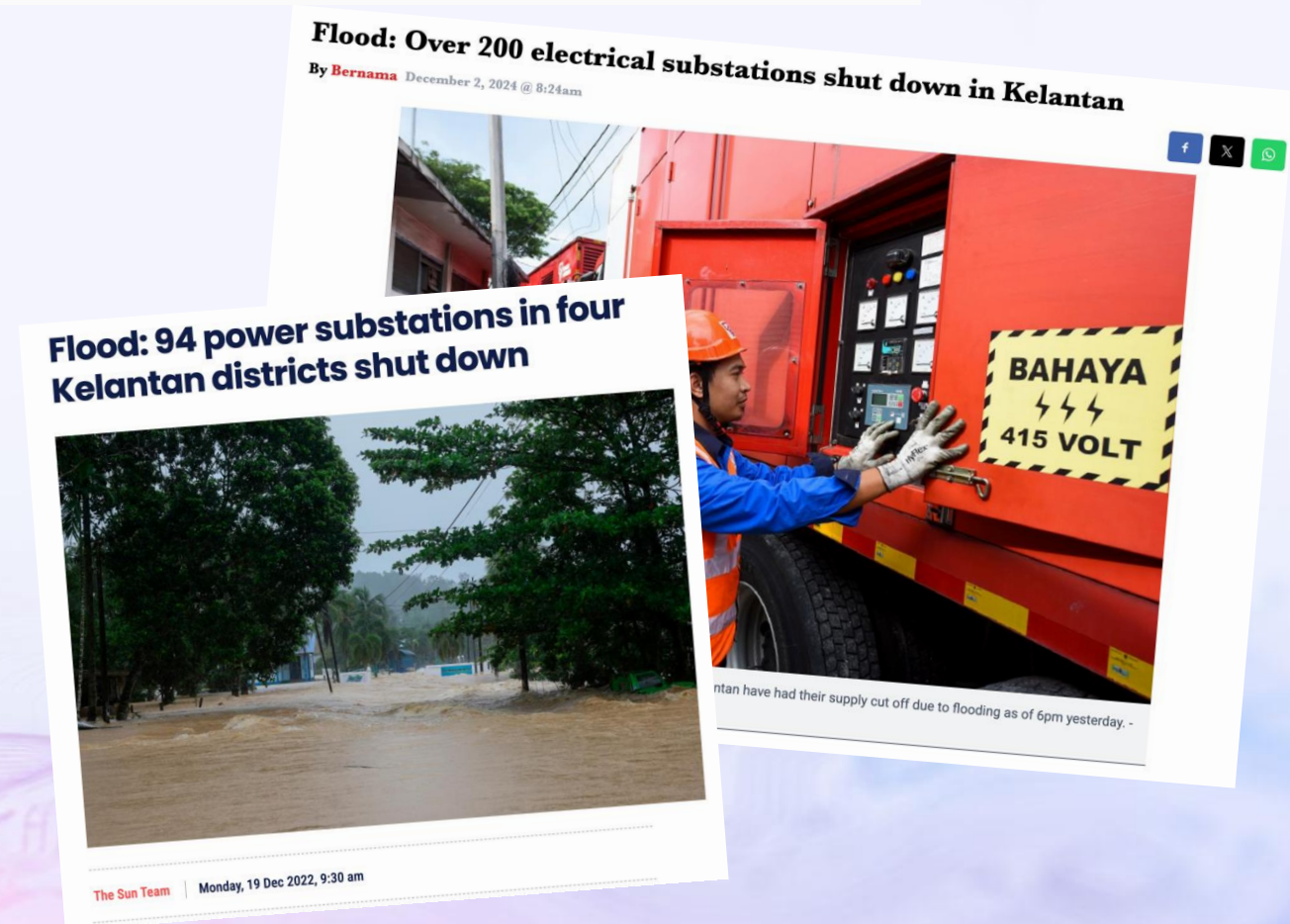
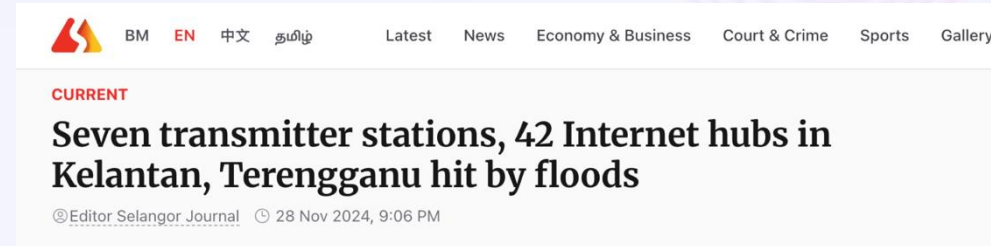
Objective

One measurable goal: keep the most people connected during flood

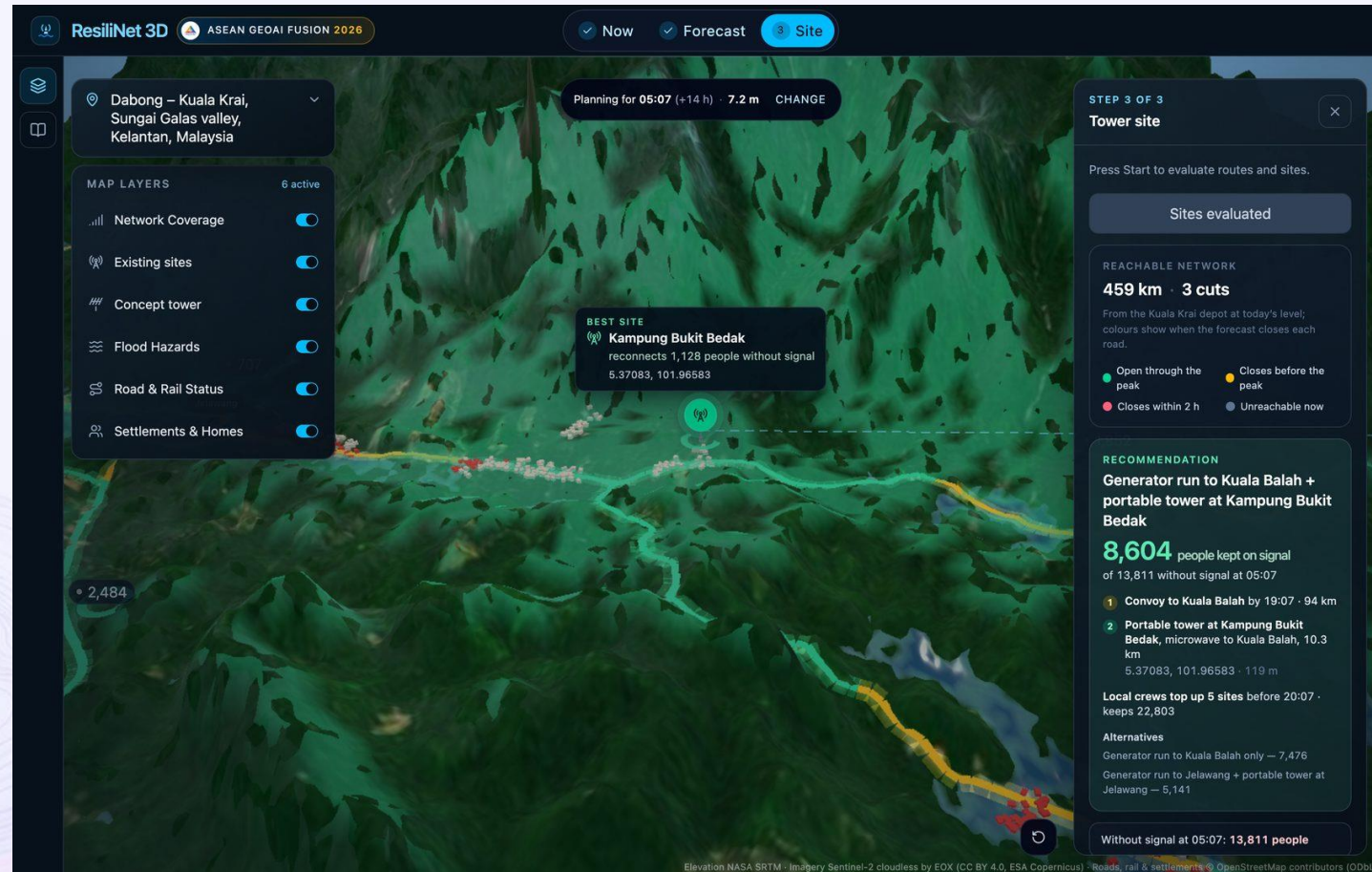
- ✓ **Main objective:** To identify where to deploy one portable tower and one generator convoy during a flood so the greatest number of people keep mobile signal.
- ✓ **Specific goals:** Predict which sites fail and when; map which roads stay open and until what hour; rank candidate sites by people reconnected.
- ✓ **Success measure:** People kept on signal. In our Kelantan scenario, 8,604 of the 13,811 who would otherwise lose coverage.

Problem Statement

- **Core problem:** Telecom sites fail in floods because of the power cut off and the roads needed to reach them close first.
- **Affected users:** Cut-off rural communities, and the one crew sent to restore their signal.
- **Location context:** Annual monsoon flooding in Kelantan and Sabah river valleys.



Live Demo



Solution Methodology



Data Input

NASA SRTM · Sentinel-2 ·
OpenStreetMap · WorldPop · JPS
InfoBanjir · WeatherNext 3



Data Preparation

62 m terrain grid · HAND flood
surface · junction-preserving
road graph



GeoAI / Analysis

AI rain → river level · roads
close hour by hour · Dijkstra
reach · viewshed ranks hills



Prototype Output

Interactive 3D app: flood,
road closures, failing sites,
a ranked tower with
coordinates



Validation

December 2014 replayed against
the record, misses reported ·
re-run on a second valley in
Sabah

Benefit / Expected Impact

A faster decision, and one the officer can defend afterwards.



Decision support

Where to send the portable tower and generator convoy. Ranked GPS coordinates in minutes instead of a guess off a paper map.



Expected impact

8,604 of 13,811 people kept on signal in our scenario — the same one tower and one convoy, better placed.



Telecom relevance

Supports network resilience, coverage continuity and disaster response for both the regulator and the operator.

Limitation / Key Challenges

What we know is weak, and what we would do about it.

- ✓ **Data limitation:** 30 m SRTM elevation reads canopy, not ground, over forest. Site list is inferred from OpenStreetMap and OpenCellID, not an operator inventory.
- ✓ **Technical limitation:** Coverage is terrain line of sight, not full Radio Frequency propagation.
- ✓ **Validation limitation:** No field trial and no agency deployment. The 2014 hindcast checks four reported facts, not a full historical series.
- ✓ **Mitigation plan:** Connect the live JPS gauge feed, ingest operator site and battery data, and replace line of sight with a propagation model.

Thank You